

Photosynthesis Float-Off Playoffs

Make leaf disks float by producing oxygen.

Win condition: Fastest half-float time with controlled variables.

THE SCIENCE YOU ARE USING

Photosynthesis makes oxygen. Leaf disks sink when the air is drawn out of them. In light, with carbon dioxide available from a baking-soda solution, the leaf photosynthesizes and produces oxygen. The oxygen collects inside the disk until it floats. Floating is direct evidence of photosynthesis.

Controlled variables. Light level, carbon-dioxide concentration, and temperature all change the rate. You run a controlled assay, change one variable, and time how long until half the disks float.

HOW TO BUILD AND RUN IT

- 01 Cut and sink the disks.** Punch leaf disks, then use a needle-free syringe to pull the air out so they sink.
- 02 Set up the solution.** Place disks in baking-soda solution to supply carbon dioxide.
- 03 Add light.** Put the cup under a light source to drive photosynthesis.
- 04 Time the half-float.** Record the time until half the disks rise to the surface.
- 05 Change one variable.** Adjust light or carbon-dioxide level, hold the rest constant, and compare rates.

Safety: No tasting. Handle the needle-free syringes as lab tools, do not point or squirt them at people, and work over the bag-covered table and wear gloves to handle the spinach and the soapy baking-soda solution; wipe spills right away. No goggles are needed for a few drops of dilute baking-soda solution. Allergy: Spinach handling note, so wear gloves.

MATERIALS

Spinach leaves	shared
Baking soda solution	shared
Cups	1 per team (4)
Oral syringes, no needle	1 per team (4)
Timers	1 per team (4)

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE variable we changed (light distance, light brightness, or baking-soda concentration): _____

Half-float time, baseline (minutes to 5 of 10 disks floating): _____

Half-float time, after redesign: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Performance	35
Controlled-variable design	25
Data table	20
Explanation	10
Cleanup	10
Total	100